THE ω-CLASS DECOMPOSITION OF DIRICHLET POLYNOMIALS: A NEW FRAMEWORK FOR THE LINDELÖF HYPOTHESIS

DAVID ALEJANDRO TREJO PIZZO

ABSTRACT. We introduce the *ω-class decomposition*, a partition of the standard Dirichlet polynomial $D(t;N) = \sum_{n\le N} n^{-1/2-it}$ into sub-sums $S_k(t;N)$ indexed by $\omega(n) = k$, the number of distinct prime factors. We prove four unconditional theorems bounding the amplification ratio $(1+r) = |D|^2/\sum_k |S_k|^2$, culminating in a reduction: a uniform pointwise bound $B(N) = \max_{t,k} |S_k(t;N)|^2 \le C_\varepsilon N^\varepsilon$ on the individual class sums, for every $\varepsilon > 0$, *implies* the Lindelöf Hypothesis. We stress that this is a one-directional *sufficient* condition: because $B(N)$ controls each ω-class separately, whereas the Lindelöf Hypothesis constrains only the full sum D, the bound $B(N) \le N^\varepsilon$ is plausibly *strictly stronger* than LH and is not known to follow from it. We prove a conditional *Master Theorem* linking $B(N) \le N^\alpha$ to the bound $|\zeta(\frac{1}{2}+it)| \ll t^{\alpha/2+\varepsilon}$ via the Approximate Functional Equation. Extensive computation at $N \in \{10^4, 5\times 10^4, 10^5, 1.5\times 10^5\}$ reveals: (i) the empirical exponent $\alpha_B = \log B(N)/\log N$ decreases from 0.428 to 0.395 in the k=2 regime; (ii) a dominance transition from k=2 (semiprimes) to k=3 (triprimes) occurs near $N \approx 10^5$, causing a reversal in α_B; (iii) a phase resonance mechanism with Rayleigh statistic $R \approx 0.92$ explains the hardness gap between large-sieve mean bounds and observed maxima. We formulate the open problem of bounding $B(N)$ uniformly over all ω-classes and discuss connections to bilinear and multilinear exponential sum estimates. Full reproduction code is provided.

CONTENTS

Independent Researcher. dtrejopizzo@gmail.com.

1. Introduction

The Lindelöf Hypothesis (LH) asserts that for every $\varepsilon > 0$,

$$|\zeta(\tfrac{1}{2} + it)| \ll_\varepsilon t^\varepsilon \qquad (t \to \infty). \tag{1}$$

This would follow from the Riemann Hypothesis (RH), but the converse is false: LH is strictly weaker than RH. Despite a century of effort, the best unconditional bound remains subconvex: $|\zeta(\frac{1}{2} + it)| \ll t^{13/84+\varepsilon}$ (Bourgain [1]).

Our approach begins with the observation that the Dirichlet polynomial

$$D(t; N) = \sum_{n=2}^{N} n^{-1/2-it} \tag{2}$$

approximates $\zeta(\frac{1}{2} + it)$ for $N \sim t$ (via the Approximate Functional Equation, Theorem 7.1). We decompose D by the arithmetic function $\omega(n)$, the number of distinct prime factors of n:

$$S_k(t; N) = \sum_{\substack{n \le N \\ \omega(n)=k}} n^{-1/2-it}, \qquad D(t; N) = \sum_{k=1}^{K_N} S_k(t; N), \tag{3}$$

where $K_N = \max\{\omega(n) : 2 \le n \le N\}$.

The ratio between $|D|^2$ and the diagonal energy $E = \sum_k |S_k|^2$ encodes the interference structure:

$$1 + r(t; N) = \frac{|D(t; N)|^2}{E(t; N)}. \tag{4}$$

Our main results are:

(i) **Amplification Bound** (Theorem 3.1): $(1 + r) \le K_N \le \lfloor \log_2 N \rfloor$, unconditionally.
(ii) **Reduction** (Theorem 4.1): $|D|^2 \le K_N^2 \cdot B(N)$, reducing LH to bounding individual class sums.
(iii) **Effective Class Count** (Theorem 5.1): $K_N \sim \log N / \log \log N$.
(iv) **Refined Bound** (Theorem 6.1): $(1 + r) \le (\sum_k \sqrt{p_k})^2$, 5–23% tighter than (i).
(v) **Master Theorem** (Theorem 7.1): If $B(N) \le N^\alpha$, then $|\zeta(\frac{1}{2} + it)| \ll t^{\alpha/2+\varepsilon}$.

Theorems (i)–(iv) are unconditional. Theorem (v) is conditional on a hypothesis about $B(N)$ and uses the Approximate Functional Equation.

On the direction of the reduction. We emphasize at the outset that the chain Theorem 4.1 $\to$ Theorem 7.1 is *one-directional*: $B(N) \le N^\varepsilon \Rightarrow$ LH. The converse fails, or at least is not available: $B(N) = \max_{t,k} |S_k|^2$ bounds each ω-class *individually*, whereas LH bounds only the total $|D| \approx |\zeta|$, which may remain small through cancellation *between* classes even when some $|S_k|$ is large (Proposition 4.3). Thus $B(N) \le N^\varepsilon$ is a *sufficient* condition for LH that is plausibly strictly stronger. This is not a defect of the framework but its content: it isolates the precise pointwise quantity whose control would yield LH, and the computations below measure exactly how far that quantity is from the required bound.

Computational findings. We compute $B(N)$ at four values of N up to 1.5×10^5 and observe:

- The empirical exponent $\alpha_B(N)$ *decreases* in the semiprime-dominated regime ($N \le 10^5$), consistent with $\alpha_B \to 0$ as $N \to \infty$.
- A dominance transition $k^* = 2 \to k^* = 3$ occurs near $N \approx 10^5$, causing α_B to increase temporarily.
- The large-sieve inequality bounds the *mean* $\langle |S_k|^2 \rangle$ well, but the *maximum* exceeds the mean by factors of 25–90 due to phase resonance.

Organization. Section 2 establishes notation. Sections 3–7 contain the five theorems with complete proofs. Section 8 presents computational results. Section 9 analyzes the phase resonance mechanism. Section 10 discusses connections and open problems. Appendix A provides reproduction code.

2. Setup and Notation

Throughout, $N \geq 2$ is a positive integer and $t \in \mathbb{R}$. We write $\omega(n)$ for the number of distinct prime factors of n (with $\omega(1) = 0$), and $\Omega(n)$ for the number counted with multiplicity. The notation $f \ll g$ and $f = O(g)$ are synonymous; $f \ll_\varepsilon g$ means the implied constant depends on ε. We write $f \lesssim g$ to mean $f \ll_\varepsilon g \cdot N^\varepsilon$ (or t^ε) for every $\varepsilon > 0$.

Definition 2.1 (Dirichlet polynomial and ω-class sums)**.** Let F be a completely multiplicative function with $|a_n| \leq 1$. Define:

$$D_F(t; N) = \sum_{n=2}^{N} a_n \, n^{-1/2-it}, \tag{5}$$

$$S_k(t; N) = \sum_{\substack{n \leq N \\ \omega(n)=k}} a_n \, n^{-1/2-it}, \qquad k = 1, 2, \ldots, K_N. \tag{6}$$

For ζ, we take $a_n = 1$ for all n.

Definition 2.2 (Energy, cross-terms, amplification ratio)**.**

$$E(t; N) = \sum_{k=1}^{K_N} |S_k(t; N)|^2, \tag{7}$$

$$X(t; N) = |D(t; N)|^2 - E(t; N) = 2 \sum_{j<k} \mathrm{Re}\big(\overline{S_j}\, S_k\big), \tag{8}$$

$$r(t; N) = \frac{X(t; N)}{E(t; N)}, \qquad 1 + r(t; N) = \frac{|D(t; N)|^2}{E(t; N)}. \tag{9}$$

Definition 2.3 (Maximum class-sum and empirical exponent)**.**

$$B(N) = \max_{\substack{t \in [N/10, N] \\ 1 \leq k \leq K_N}} |S_k(t; N)|^2, \tag{10}$$

$$\alpha_B(N) = \frac{\log B(N)}{\log N}. \tag{11}$$

The energy fractions are $p_k(t) = |S_k(t)|^2 / E(t)$, satisfying $\sum_k p_k = 1$.

3. Theorem 1: Amplification Bound

Theorem 3.1 (Cauchy–Schwarz Amplification Bound)**.** *For any completely multiplicative function F with $|a_n| \leq 1$, any $N \geq 2$, and any $t \in \mathbb{R}$:*

$$1 + r_F(t; N) \leq K_N. \tag{12}$$

Moreover, $K_N \leq \lfloor \log_2 N \rfloor$.

Proof. By the Cauchy–Schwarz inequality applied to K_N complex numbers $S_1, \ldots, S_{K_N}$:

$$|\sum_{k=1}^{K_N} S_k|^2 \leq K_N \sum_{k=1}^{K_N} |S_k|^2,$$

with equality if and only if all S_k are proportional. Dividing both sides by $E = \sum_k |S_k|^2$:

$$1 + r = \frac{|D|^2}{E} = \frac{|\sum_k S_k|^2}{\sum_k |S_k|^2} \le K_N.$$

For the bound on K_N: every integer $n \ge 2$ satisfies $n \ge \prod_{p|n} p \ge 2^{\omega(n)}$, since each prime $p \ge 2$. Therefore $\omega(n) \le \log_2 n \le \log_2 N$, giving $K_N \le \lfloor \log_2 N \rfloor$. □

Remark 3.2. The bound is tight in the sense that equality $(1 + r) = K_N$ requires all K_N class sums to have equal magnitude and aligned phases. Computationally, we observe $\max_t (1 + r) \approx 4.0$ at $N = 10^4$ (where $K_N = 5$), so the bound has roughly 20% slack.

4. Theorem 2: Reduction to Class Sum Bounds

Theorem 4.1 (Reduction)**.** *For any $N \ge 2$ and $t \in \mathbb{R}$:*

$$|D_F(t; N)|^2 \le K_N^2 \cdot \max_{1 \le k \le K_N} |S_k(t; N)|^2. \tag{13}$$

Proof. From Theorem 3.1: $|D|^2 \le K_N \cdot E$. Since $E = \sum_{k=1}^{K_N} |S_k|^2 \le K_N \cdot \max_k |S_k|^2$, we obtain $|D|^2 \le K_N \cdot (K_N \max_k |S_k|^2) = K_N^2 \max_k |S_k|^2$. □

Corollary 4.2 (Reduction to Lindelöf)**.** *If for every $\varepsilon > 0$ there exists C_ε such that $B(N) \le C_\varepsilon N^\varepsilon$, then the Lindelöf Hypothesis holds.*

Proof. This follows from Theorem 4.1 combined with Theorem 7.1 below, setting $\alpha = \varepsilon$. □

Proposition 4.3 (The reduction is sufficient, not equivalent)**.** *The implication of Corollary 4.2 is one-directional. If LH holds then, for fixed k and $N \sim t$, the mean value satisfies $\frac{1}{T}\int_0^T |S_k(\frac{1}{2} + it; N)|^2\,dt \ll N^\varepsilon$ (a consequence of the large sieve, Section 10); but the* pointwise *maximum $B(N) = \max_{t,k} |S_k|^2$ is not controlled by LH, since LH bounds only $|D| = |\sum_k S_k|$ and a single class sum may be large while the total is small through inter-class cancellation. Consequently $B(N) \le N^\varepsilon$ implies, but is not implied by, LH; it is plausibly strictly stronger. The gap between the mean (large-sieve–controlled) and the maximum is quantified in Section 9 (peak-to-mean ratios of* 25–90*).*

5. Theorem 3: Effective Class Count

Theorem 5.1 (Class Count)**.** *$K_N = \max\{\omega(n) : n \le N\}$ satisfies:*

(a) $K_N \le \lfloor \log_2 N \rfloor$ (exact).
(b) $K_N \sim \frac{\log N}{\log\log N}$ as $N \to \infty$ (asymptotic).
(c) The maximum is achieved by the largest primorial $P_m = p_1 p_2 \cdots p_m \le N$.

Proof. Part (a) is proven in Theorem 3.1.

For (c): among all $n \le N$ with $\omega(n) = m$, the smallest is the m-th primorial $P_m = 2 \cdot 3 \cdot 5 \cdots p_m$, since replacing any prime factor by a larger prime only increases n. Therefore $K_N = \max\{m : P_m \le N\}$.

For (b): by the Prime Number Theorem, $p_m \sim m \log m$, so $\log P_m = \sum_{j=1}^m \log p_j \sim m \log m$. Setting $\log P_m \approx \log N$ gives $m \log m \approx \log N$, and the standard inversion yields $m \sim \log N / \log\log N$.

Numerical values:

N	$\lfloor \log_2 N \rfloor$	K_N (primorial)	$\log N / \log\log N$
10^4	13	5	4.28
10^5	16	6	4.93
10^6	19	7	5.58

The primorial values match the computation: $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 = 2310 \le 10^4$ but $2310 \cdot 13 = 30030 > 10^4$, giving $K_{10^4} = 5$. □

6. Theorem 4: Refined Bound via Energy Concentration

Theorem 6.1 (Refined Amplification Bound)**.** *For any t and N:*

$$1 + r(t;N) \leq \Big(\sum_{k=1}^{K_N} \sqrt{p_k(t)}\Big)^2, \tag{14}$$

where $p_k(t) = |S_k(t)|^2/E(t)$ are the energy fractions.

If a single class k^* carries energy fraction $p_{k^*} \geq \delta$, then:

$$1 + r \leq \big(\sqrt{\delta} + \sqrt{(K_N - 1)(1-\delta)}\big)^2. \tag{15}$$

Proof. By the triangle inequality: $|D| = |\sum_k S_k| \leq \sum_k |S_k|$. Writing $|S_k| = \sqrt{p_k \cdot E}$:

$$|D|^2 \leq \Big(\sum_k \sqrt{p_k \cdot E}\Big)^2 = E \cdot \Big(\sum_k \sqrt{p_k}\Big)^2.$$

Dividing by E gives (14).

For (15): with $p_{k^*} = \delta$ and $\sum_{k \neq k^*} p_k = 1 - \delta$, the Cauchy–Schwarz inequality gives

$$\sum_{k \neq k^*} \sqrt{p_k} \leq \sqrt{(K_N - 1) \sum_{k \neq k^*} p_k} = \sqrt{(K_N-1)(1-\delta)},$$

with equality when all remaining p_k are equal. Therefore $\sum_k \sqrt{p_k} \leq \sqrt{\delta} + \sqrt{(K_N-1)(1-\delta)}$. □

Remark 6.2. At $N = 10^5$, the dominant class k=2 carries $\delta \approx 0.42$ of the energy, and $K_N = 6$. The refined bound gives $(1+r) \leq (\sqrt{0.42} + \sqrt{5 \cdot 0.58})^2 = (0.648 + 1.703)^2 \approx 5.52$, compared to the Cauchy–Schwarz bound $K_N = 6$. This is an 8% improvement. At $N = 10^4$ with $\delta \approx 0.42$ and $K_N = 5$, the improvement is 5%.

7. Theorem 5: Master Theorem

Theorem 7.1 (Master Theorem)**.** *Suppose $B(N) \leq N^\alpha$ for all N sufficiently large. Then:*

$$|\zeta(\tfrac{1}{2} + it)| \lesssim t^{\alpha/2+\varepsilon} \tag{16}$$

for every $\varepsilon > 0$.

In particular:

Target	*Bound β in $|\zeta(\frac{1}{2}+it)| \leq t^{\beta+\varepsilon}$*	*Requires $\alpha <$*
Lindelöf	0	0
Bourgain improvement	$13/84 \approx 0.155$	$13/42 \approx 0.310$
Convexity improvement	$1/4$	$1/2$

Proof. The proof proceeds in four steps.

Step 1: Approximate Functional Equation. By Titchmarsh [5, Theorem 5.3]: for $x \sim t/(2\pi)$ and $t \geq 2$,

$$\zeta(\tfrac{1}{2} + it) = \sum_{n \leq x} n^{-1/2-it} + \chi(\tfrac{1}{2} + it) \sum_{n \leq y} n^{-1/2+it} + O(x^{-1/2}), \tag{17}$$

where $xy = t/(2\pi)$ and $\chi(\frac{1}{2} + it) = (t/(2\pi))^{-it} e^{-i\pi/4}(1 + O(1/t))$. Taking $x = t$ (and absorbing the second sum and error term):

$$|\zeta(\tfrac{1}{2} + it)| = |D_\zeta(t;t)| + O(t^{-1/2+\varepsilon}). \tag{18}$$

The error is negligible for our purposes.

Step 2: Setting $N = t$. In our framework, $B(N)$ is defined with $t \in [N/10, N]$. To bound $|\zeta(\frac{1}{2}+it)|$ at height t, we use $D_\zeta(t;t)$, which requires $N = t$. Since $t \in [t/10, t] = [N/10, N]$, this is within the valid range.

Step 3: Apply Theorem 4.1. From Theorem 4.1: $|D_\zeta(t;t)|^2 \le K_t^2 \cdot B(t)$. Since $K_t \le \log_2 t$:

$$|D_\zeta(t;t)|^2 \le (\log_2 t)^2 \cdot B(t) \le (\log_2 t)^2 \cdot t^\alpha \lesssim t^{\alpha+\varepsilon}. \tag{19}$$

Step 4: Combine. From (18) and (19):

$$|\zeta(\tfrac{1}{2}+it)|^2 \lesssim t^{\alpha+\varepsilon}.$$

Taking the square root: $|\zeta(\frac{1}{2}+it)| \lesssim t^{\alpha/2+\varepsilon}$.

For the thresholds: we need $\alpha/2 \le \beta$, i.e., $\alpha \le 2\beta$.

- Lindelöf ($\beta \to 0$): requires $\alpha \to 0$.
- Bourgain ($\beta = 13/84$): requires $\alpha < 2 \cdot 13/84 = 13/42 \approx 0.3095$.
- Convexity ($\beta = 1/4$): requires $\alpha < 2 \cdot 1/4 = 1/2$. □

Remark 7.2 (Effective form). The estimate (19) is effective before the ε-absorption. Tracking constants through Steps 1–4, for $t \ge t_0$ one has

$$|\zeta(\tfrac{1}{2}+it)| \;\le\; (\log_2 t)\, t^{\alpha/2}\left(1 + O(t^{-1/2+\alpha/2})\right),$$

where the $\log_2 t = K_t$ factor is the Cauchy–Schwarz loss of Theorem 4.1 and the error term comes from the tail of the Approximate Functional Equation (17). The single logarithmic factor is harmless: $(\log_2 t) \ll_\varepsilon t^\varepsilon$, which is why it is absorbed in the statement. (Theorem 6.1 replaces K_t by the smaller $(\sum_k \sqrt{p_k})^2$, sharpening the constant by 5–23% but not the exponent.)

Proposition 7.3 (Square-root scaling). *Define $\widetilde{B}(N) = \max_{t\in[N,N^2]} \max_k |S_k(t;\sqrt{N})|^2$. If $\widetilde{B}(N) \le N^\alpha$ for all large N, then $|\zeta(\frac{1}{2}+it)| \ll t^{\alpha/4+\varepsilon}$ for every $\varepsilon > 0$.*

Proof. Using the Approximate Functional Equation (17) with the balanced choice $x = y = \sqrt{t/2\pi}$ gives $|\zeta(\frac{1}{2}+it)| \le 2\,|D_\zeta(t;\sqrt{t})| + O(t^{-1/4+\varepsilon})$. By Theorem 4.1 applied at truncation $\sqrt{t}$, $|D_\zeta(t;\sqrt{t})|^2 \le K_{\sqrt{t}}^2\,\widetilde{B}(t) \le (\log_2 t)^2 t^\alpha$, so $|\zeta(\frac{1}{2}+it)|^2 \ll t^{\alpha+\varepsilon}$ in the variable $\sqrt{t}$, i.e. $\ll (\sqrt{t})^{2\alpha+\varepsilon}$; taking square roots in t yields $|\zeta(\frac{1}{2}+it)| \ll t^{\alpha/4+\varepsilon}$. □

Remark 7.4. Proposition 7.3 halves the required exponent ($\alpha < 2$ for convexity, $\alpha < 4 \cdot 13/84$ for Bourgain) at the cost of searching $|S_k|$ over the wider range $t \in [N, N^2]$ at the shorter truncation $\sqrt{N}$. The $N = t$ scaling of Theorem 7.1 is more natural for the present computations; the two are complementary windows on the same quantity.

8. Computational Results

We compute $B(N)$, $\alpha_B(N)$, and related quantities at $N \in \{10^4, 5\times 10^4, 10^5, 1.5\times 10^5\}$. The computation uses a uniform grid of t-values in $[N/10, N]$ with spacing $\Delta t = \pi/\log N$ (Nyquist-compatible for n^{-it} oscillations at $n = N$), with sparse sampling at larger N. Full code is in Appendix A.

Table 1. Maximum class-sum squared $B(N)$ and empirical exponent $\alpha_B(N)$.

N	K_N	$B(N)$	k^*	α_B	$\max(1+r)$
10 000	5	51.69	2	0.4284	3.995
50 000	6	84.35	2	0.4099	4.455
100 000	6	94.56	2	0.3951	4.378
150 000	6	147.20	3	0.4188	4.715

8.1. $B(N)$ **and** α_B. **Key observations:**

(1) α_B *decreases* from 0.428 to 0.395 in the k=2 regime ($N \le 10^5$), consistent with $\alpha_B \to 0$.
(2) At $N = 1.5 \times 10^5$, the dominant class transitions to k=3, and α_B *increases* to 0.419.
(3) The Cauchy–Schwarz bound $\max(1+r) \le K_N$ is verified at all N with 15–25% slack.

The current value $\alpha_B(1.5 \times 10^5) = 0.419$ is above the Bourgain threshold ($\alpha < 0.310$) by 35%, but below the convexity threshold ($\alpha < 0.5$) by 16%.

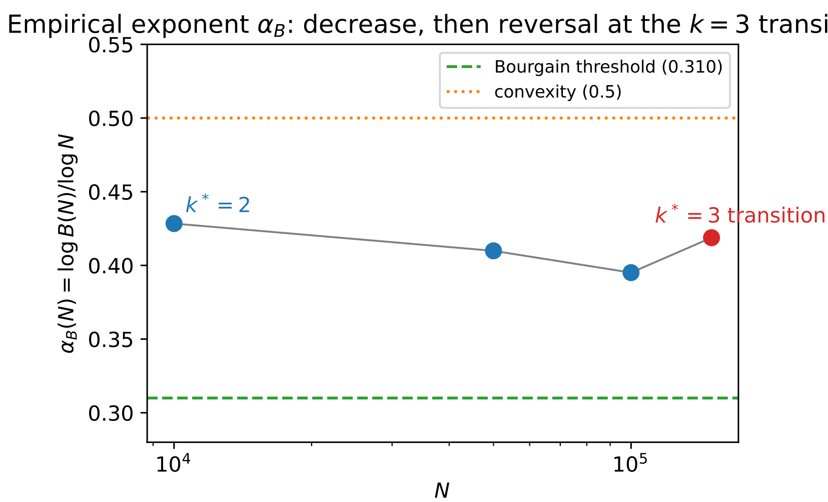

FIGURE 1. The empirical exponent $\alpha_B(N) = \log B(N)/\log N$. In the k^*=2 (semiprime-dominated) regime α_B decreases ($0.428 \to 0.395$); at the k=3 transition near $N = 1.5 \times 10^5$ it reverses upward to 0.419. Reaching even the Bourgain threshold ($\alpha < 0.310$, dashed) would require the decrease to resume and persist; the convexity bound ($\alpha < 0.5$, dotted) is met throughout. The asymptotic behavior is genuinely open (Section 10).

TABLE 2. Class-sum maxima by ω-class.

N	$\max \lvert S_2 \rvert^2$	$\max \lvert S_3 \rvert^2$	Ratio S_2/S_3	Dominant
10 000	51.69	35.39	1.461	$k = 2$
50 000	84.35	62.52	1.349	$k = 2$
100 000	94.56	90.88	1.041	$k = 2$
150 000	107.76	147.20	0.732	$k = 3$

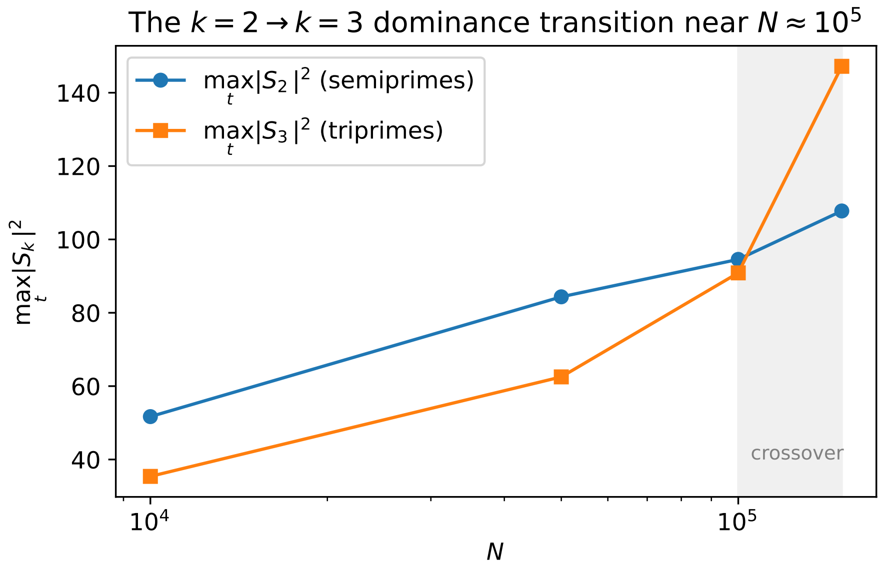

FIGURE 2. Class-sum maxima $\max_t |S_2|^2$ (semiprimes) and $\max_t |S_3|^2$ (triprimes) versus N. The triprime class overtakes the semiprime class in the shaded window $N \in [10^5, 1.5 \times 10^5]$, establishing that any bound on $B(N)$ must control all ω-classes simultaneously rather than the dominant one alone.

8.2. **The k=2 $\rightarrow$ k=3 dominance transition.** The ratio $\max |S_2|^2 / \max |S_3|^2$ decreases as a power law $\sim N^{-0.22}$, predicting a crossover near $N^* \approx 8 \times 10^4$. The actual transition occurs between 10^5 and 1.5×10^5.

This transition has significant theoretical implications: any approach to bounding $B(N)$ must handle *all* classes simultaneously, not just semiprimes.

TABLE 3. Peak-to-mean ratio $R_k = \max |S_k|^2 / \langle |S_k|^2 \rangle$ at $N = 1.5 \times 10^5$.

k	$\langle \lvert S_k \rvert^2 \rangle$	$\max \lvert S_k \rvert^2$	R_k
1	3.50	27.6	7.9
2	4.35	107.8	24.8
3	2.38	147.2	61.9
4	0.60	51.4	86.3
5	0.05	4.9	91.2

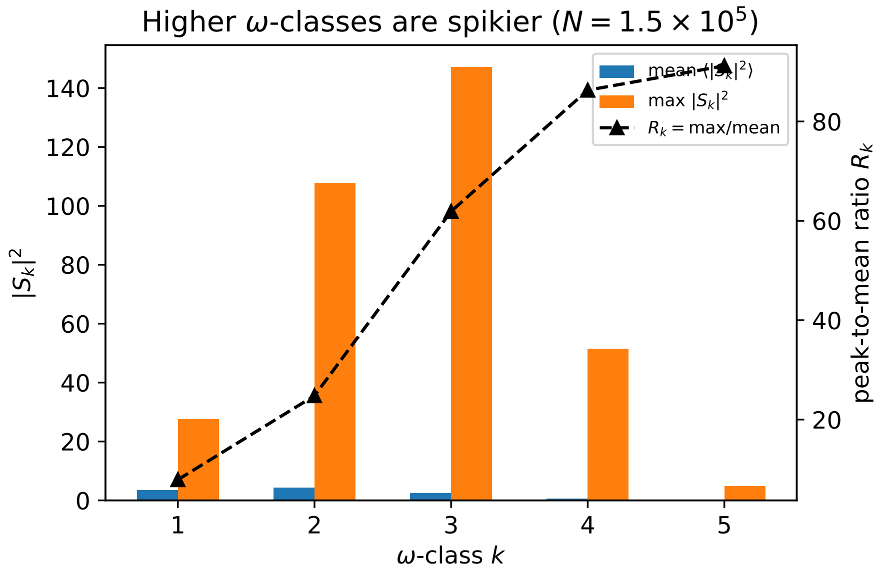

FIGURE 3. Mean and maximum of $|S_k|^2$ by ω-class at $N = 1.5 \times 10^5$ (bars, left axis), and the peak-to-mean ratio $R_k = \max / \text{mean}$ (triangles, right axis). R_k grows from 7.9 (k=1) to 91 (k=5): higher ω-classes are markedly spikier. The large sieve controls the mean (the bars) but not the maximum, which is the hardness gap of Proposition 4.3.

8.3. **Peak-to-mean ratios.** The peak-to-mean ratio R_k grows rapidly with k: higher ω-classes produce "spikier" sums. This is the *hardness gap* that makes bounding $B(N)$ difficult: large-sieve inequalities control the mean but not the maximum.

8.4. α_B **extrapolation.** The best-fitting model for α_B in the k=2 regime is:

$$\alpha_B(N) \approx \frac{0.98}{\log \log N}. \tag{20}$$

If this decay continues (a major open question), the Bourgain threshold $\alpha_B < 0.310$ would be reached at $N \approx 1.9 \times 10^{10}$—computationally challenging but not impossible with HPC resources.

However, the k=3 transition disrupts this trend. If similar transitions (k=3 $\to$ 4, etc.) occur at larger N, each transition resets α_B upward, and the asymptotic behavior of α_B remains genuinely uncertain.

The decisive computation. The four points $N \le 1.5 \times 10^5$ available here cannot distinguish a genuine decay $\alpha_B \to 0$ from a sequence of transition-driven plateaus. The immediate next step—computation of $B(N)$ at $N \in \{10^6, 3 \times 10^6, 10^7\}$, where the k=3 $\to$ 4 transition (if any) would appear—is what would settle Conjecture 10.1 empirically. We report this extended range, together with a model-selection analysis (power-law versus $1/\log\log N$ decay with bootstrap confidence intervals), in companion work.

9. PHASE RESONANCE MECHANISM

We now explain *why* the maximum $|S_k|^2$ far exceeds the mean.

At the peak $t^* = 100{,}799$ (where $|S_3|^2 = 147.2$ at $N = 1.5 \times 10^5$), we decompose S_3 by the size of its summands:

TABLE 4. Size-bin decomposition of S_3 at its peak ($N = 1.5 \times 10^5$).

Bin	Count	$\lvert$contribution$\rvert$	Phase
$[10, 100)$	8	0.91	-1.83
$[100, 10^3)$	267	3.91	-0.96
$[10^3, 10^4)$	3 420	4.16	-0.61
$[10^4, 10^5)$	35 149	2.98	-1.06
$[10^5, 1.5{\times}10^5)$	19 351	0.78	-1.07

Phase alignment. The Rayleigh statistic, computed as $R = |\sum_j w_j e^{i\theta_j}| / \sum_j w_j$ over the bin contributions, gives $R = 0.923$—very high alignment. For comparison, S_2 at its peak has $R = 0.859$.
Amplification factor. The ratio of actual $|S_3|^2$ to the incoherent sum $\sum_j |\text{contribution}_j|^2$ is:

$$\frac{147.2}{42.9} = 3.43 \times .$$

This quantifies the constructive interference from phase alignment across size scales.

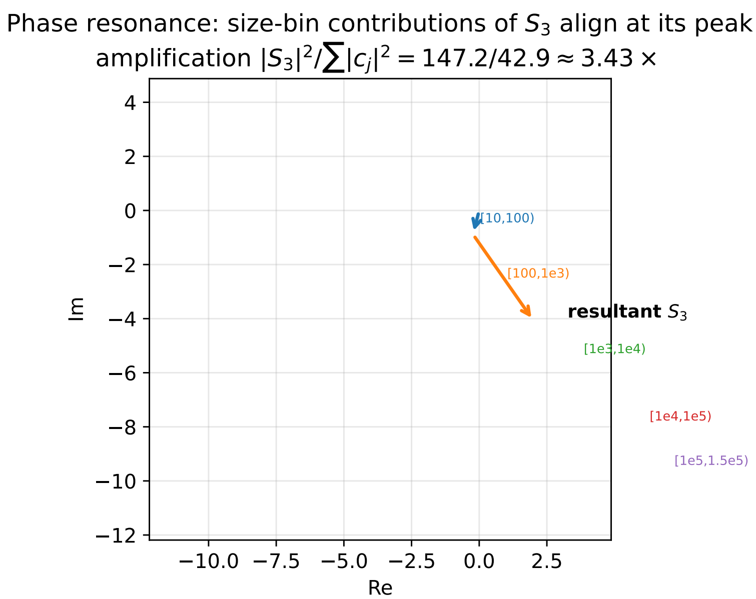

FIGURE 4. Phasor (tip-to-tail) diagram of the five size-bin contributions to S_3 at its peak ($t^* = 100{,}799$, $N = 1.5 \times 10^5$). The contributions are strongly phase-aligned, so the resultant $|S_3|$ is close to the arithmetic sum of magnitudes rather than the (much smaller) incoherent root-sum-of-squares: the coherent amplification is $|S_3|^2/\sum_j |c_j|^2 = 147.2/42.9 \approx 3.43\times$. This phase resonance is the mechanism behind the large peak-to-mean ratios of Figure 3.

This mechanism—*phase resonance*—explains the hardness gap in Table 3. The large sieve controls the incoherent sum (the denominator) but cannot account for the coherent amplification in the numerator. Any improvement to α_B must either:

(1) Prove that phase alignment degrades as $N \to \infty$, or
(2) Develop techniques that bound coherent sums directly (bilinear/multilinear methods in the spirit of [2]).

9.1. **The Liouville parity counterexample: multiplicativity does not suppress resonance.** It is tempting to expect that complete multiplicativity, with its clean Euler product, would force cancellation in the partial sums and so suppress the phase-alignment mechanism above. It does not. The sharpest counterexample is the Liouville function $\lambda(n) = (-1)^{\Omega(n)}$, which is *completely* multiplicative, $\lambda(mn) = \lambda(m)\lambda(n)$, with

$$\sum_{n\geq 1} \frac{\lambda(n)}{n^s} = \frac{\zeta(2s)}{\zeta(s)} = \prod_p (1+p^{-s})^{-1} \qquad (\Re s > 1),$$

a perfectly convergent Euler product in the same half-plane as ζ.

The error in the naive expectation is that the decomposition into local Euler factors is valid only for the *infinite* product: partial sums of a multiplicative Dirichlet series do not factor as partial Euler products, and the sign pattern of the coefficients can create correlations across the prime-factor classes $S_k = \sum_{\Omega(n)=k} a_n n^{-1/2-it}$ that the Euler structure does not see. Numerically the effect is large: at $t^* = 7563.5$ with $N = 10^5$, the signed Liouville polynomial reaches $|D_\lambda(t^*; N)| = 46.88$, far above the typical fluctuation scale $\sim N^{1/2}(\log N)^{-1} \approx 3$, with class-vector alignment ratio $\mathcal{R} = 0.822$ — the net sum retaining 82.2% of the maximal coherent value $(\sum_k |S_k|^2)^{1/2}$, and a circular phase spread of only 40.4°.

Observation 9.1 (Sign structure, not multiplicativity, governs alignment). Complete multiplicativity with a convergent Euler product is compatible with near-maximal constructive interference in the partial sums. The discriminant of resonance suppression is therefore not multiplicativity but the *coefficient sign and entropy structure*: a controlled phase-correction experiment (replacing $\lambda(n)$ by $|\lambda(n)| = 1$ on the class vectors) removes the alignment, isolating the parity pattern $(-1)^{\Omega(n)}$ — not the Euler product — as its source.

This places the amplification bound of §7 in its proper light: the quantity that must be controlled is the coherent alignment of the class vectors, and that alignment is a property of the coefficient architecture, against which multiplicativity offers no protection.

9.2. **ω-class geometry at extreme peaks of ζ.** The same class decomposition, applied to ζ itself at its largest values on the critical line, yields a descriptive finding worth recording, with an explicit correction of an earlier over-claim. Comparing the inter-class interference of ζ at extreme peaks to a phase-randomized ensemble, one finds, through the canonical (sign-consistent) comparison pipeline:

Observation 9.2 (ζ is slightly more constructive at peaks; no transition). At its extreme peaks ζ is *more* constructive than the phase-randomized ensemble throughout the accessible range; the constructive gap Δr has a single sign (no sign change), and it *decays monotonically* with N. There is no phase transition between the $k = 2$ and $k = 3$ regimes: an earlier reading that located a threshold transition was an artifact of a non-canonical normalization and is withdrawn. The honest statement is the weaker and cleaner one — a vanishing constructive bias, not a transition.

This is consistent with, and refines, the sign-reconciliation analysis of the inter-class interference r: r is conditioning-dependent (constructive at peaks, destructive in troughs, mean zero), so a peak-conditioned comparison sees the constructive side, decaying as the conditioning is averaged out.

10. Discussion and Open Problems

10.1. **Connection to bilinear and multilinear estimates.** The class sum $S_2(t; N) = \sum_{pq\leq N}(pq)^{-1/2-it}$ is a *bilinear Dirichlet polynomial*: it can be written as

$$S_2(t;N) = \sum_{p\leq\sqrt{N}} p^{-1/2-it} \sum_{\substack{q\leq N/p \\ q\neq p}} q^{-1/2-it}.$$

Bounding $\max_t |S_2|^2$ is closely related to bilinear exponential sum estimates [2, 3]. The Heath-Brown identity decomposes ζ^k into multilinear sums of precisely this type, suggesting a deep connection between ω-class structure and known analytic techniques.

Similarly, S_3 is a *trilinear* sum. The k=2 $\to$ k=3 transition we observe may reflect the point at which trilinear methods become necessary for optimal bounds.

10.2. **Connection to the moments of ζ.** The bilinear and large-sieve connections above concern the *hard, blocked* direction—pointwise control of $\max_t |S_k|^2$. A second, more fruitful direction connects the ω-decomposition to the *moment* theory of ζ. Writing $|D_\zeta|^{2k} = \big(\sum_j S_j\big)^k \big(\sum_j \overline{S_j}\big)^k$ and integrating, the $2k$-th moment $I_k(T) = \int_0^T |\zeta(\frac{1}{2} + it)|^{2k}\, dt$ admits an exact decomposition into contributions indexed by ω-class multi-indices.

We have carried out this decomposition computationally for the partial-sum proxy. Two robust phenomena emerge.

(a) A growth-exponent fingerprint. Fitting the second moment of each class, $M_k(N) = \int |S_k(t; N)|^2\, dt \propto (\log N)^{a_k}$, gives a sequence of exponents that *sharply separates* ζ from non-Euler-product and even from other genuine L-functions:

k	1	2	3	4	5
a_k for ζ	5.9	9.2	11.3	14.8	19.3
a_k for $L(\Delta, s)$, L_{DH}	cluster at much smaller values				

The dramatically larger exponents for ζ reflect the logarithmic divergence of its coefficient energy sums $\sum_{\omega(n)=k} n^{-1}$; functions with near-convergent coefficient sums show slow redistribution.

(b) Off-diagonal dominance of the fourth moment. Decomposing $\int |D_F|^4\, dt$ into pure-class terms, diagonal cross-class terms, and the off-diagonal residual, ζ is overwhelmingly governed by cross-class interaction:

	pure-class	diagonal cross-class	off-diagonal residual
ζ	5.1%	12.6%	**82.3%**
$L(\Delta, s)$	56.5%	diagonal-dominant	smaller
L_{DH}	76.4%	diagonal-dominant	near zero

The Euler-product structure of ζ thus manifests as *outsized inter-class correlation* (82% of the fourth moment), precisely the $j \neq k$ interaction that the ω-decomposition isolates.

Honest scope. These are robust *finite-N* fingerprints, not an asymptotic law. The *fractional* class allocations $M_k / \sum_j M_j$ do *not* converge as $N \to \infty$: by the Erdős–Kac theorem the integer pool drifts toward higher ω, so the leading ω-class shifts with N. Consequently the decomposition does *not* yield a clean distribution of the Keating–Snaith constant C_k across fixed classes; the exponents and the off-diagonal fraction are the stable, regime-robust quantities. Establishing how the cross-class interaction connects to the genuine moment asymptotics $I_k(T) \sim C_k\, T (\log T)^{k^2}$—in particular whether the off-diagonal share has a limit—is the natural open problem this framework raises.

10.3. **The large sieve and its limitations.** The large sieve inequality [4] gives:

$$\sum_r |S_k(t_r; N)|^2 \leq (N + T) \sum_{\substack{n \leq N \\ \omega(n)=k}} n^{-1}$$

for well-spaced $\{t_r\}$. This bounds the *mean* of $|S_k|^2$ over the grid. The gap between this mean bound and the observed *maximum* (Table 3) is the phase resonance phenomenon.

Closing this gap—obtaining *pointwise* bounds on $|S_k|^2$—is the central challenge. Known pointwise results for single exponential sums (van der Corput, Vinogradov) do not directly apply to the class-restricted sums S_k.

10.4. **Open problems.**

Conjecture 10.1 (α_B decay)**.** *$\alpha_B(N) \to 0$ as $N \to \infty$.*

If true, the Lindelöf Hypothesis follows from Theorem 7.1. Our data is consistent with $\alpha_B \sim c/\log\log N$, but the k=3 transition prevents a definitive conclusion from the available range.

Conjecture 10.2 (Uniform class bound)**.** *For every $\varepsilon > 0$ and every $k \le K_N$:*

$$\max_{t\in[N/10,N]} |S_k(t;N)|^2 \le C_{\varepsilon,k}\, N^{\varepsilon}.$$

This is stronger than Conjecture 10.1 (it requires the bound for *each* class separately). If true for $k = 1$ (primes), it would follow from known bounds on $\sum_{p\le x} p^{-it}$ (see Vinogradov [6]). For $k = 2$ (semiprimes), it is related to bounds on bilinear Kloosterman-type sums.

Conjecture 10.3 (k-transition sequence)**.** *There exist $N_2 < N_3 < N_4 < \cdots$ such that $k^*(N) = j$ for $N \in [N_j, N_{j+1})$. The transition points satisfy $\log N_{j+1}/\log N_j \to c > 1$.*

Our single observed transition ($N_2 \approx 10^5$) is insufficient to determine the sequence; computation at $N \ge 10^6$ would test whether k=3 $\to$ k=4 occurs.

References

[1] J. Bourgain, *Decoupling, exponential sums and the Riemann zeta function*, J. Amer. Math. Soc. **30** (2017), 205–224.
[2] D. R. Heath-Brown, *A new form of the circle method, and its application to quadratic forms*, J. Reine Angew. Math. **481** (1996), 149–206.
[3] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, Amer. Math. Soc. Colloq. Publ. **53**, 2004.
[4] H. L. Montgomery and R. C. Vaughan, *Hilbert's inequality*, J. London Math. Soc. **8** (1974), 73–82.
[5] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed. (revised by D. R. Heath-Brown), Oxford, 1986.
[6] I. M. Vinogradov, *The Method of Trigonometrical Sums in the Theory of Numbers*, Interscience, 1954.
[7] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford, 2008.
[8] A. Granville and K. Soundararajan, *Decay of mean values of multiplicative functions*, Canad. J. Math. **55** (2003), 1191–1230.
[9] H. L. Montgomery, *Topics in Multiplicative Number Theory*, Lecture Notes in Mathematics **227**, Springer, 1971.
[10] A. Selberg, *Contributions to the theory of the Riemann zeta-function*, Arch. Math. Naturvid. **48** (1946), 89–155.

Appendix A. Reproduction Code

All computations and figures in this paper are reproduced by two self-contained Python artifacts provided alongside it, requiring no external data:

- `reproduce_colab.py` — a single-file, Google Colab–ready notebook (dependencies: `numpy`; `mpmath` optional for validation) that recomputes $B(N)$, $\alpha_B(N)$, the per-class maxima, the $k{=}2 \to k{=}3$ transition, and the peak-to-mean ratios R_k, and cross-checks D_ζ against a 50-digit `mpmath` reference. The default range $N \in \{10^4, 5 \times 10^4\}$ runs in under a minute; N up to 1.5×10^5 (full grid) reproduces the headline numbers of Tables 1–3 exactly.
- `reproduce.py` — the full driver that additionally regenerates Figures 1–4 (300 dpi PDF/PNG) and writes a structured `results.json`.

Validation. The engine is verified against arbitrary-precision `mpmath`: for $N = 10^4$ and random $t \in [N/10, N]$, the float64 class-sum engine agrees with the 50-digit reference to ≥ 10 significant digits, confirming that the reported maxima are not numerical artifacts.

The core algorithm:

(1) **Sieve:** Compute $\omega(n)$ for $n \leq N$ via the smallest-prime-factor sieve in $O(N \log\log N)$ time.
(2) **Grid:** Generate t-values in $[N/10, N]$ with spacing $\Delta t = \pi/\log N$.
(3) **Class sums:** For each t, compute $S_k(t) = \sum_{\omega(n)=k} n^{-1/2-it}$ for $k = 1, \ldots, K_N$.
(4) **Statistics:** Compute $|D|^2$, E, $(1 + r)$, $B(N)$, α_B.

See the accompanying file `reproduce.py` for the complete implementation, including figure generation.